b to their difference 21	1:32222	
So is the tangent of half the fum of?	84	1.92428
the angle ID and C	610 58’ 10.27372	
To the tangent of half their difference 79 50’	9.13806	
So by theorem, we have the angles D and C thus;		
To hold the fun of the angles I D and C	611 58’	
Add half their difference	70 50’	
And the fum is the greater angle D	69o 48’	
Alfo, from half the fum	61o 58’	
Take half the difference	70 50’	
And there remains the lifter angle C	54°08’	
Then by Cafel II, we have the following analogy for find ing On the leg required, viz.	S.C. B D :: S, B. D C.	
I. As the fine of C	54°03’ 9.90860	
To BD	133	2.12385
So is the fine of B	56°03’ 9.91883	
TOO	150A 1.13590	
Case VI. Three files being given, to find the angles.		
Examle: In the triangle A B C ((bit a 9)) fuppfe AB=13, AC=185 7, and BC=B4 y required the angles		
A, B, and C; To the triangle will be confucted, and the angles may be incurred by a line of clouds.		
II. By undertaking: Let fall the perpendicular, BD from the vertex B, upon the bafe’AC; which will divide the one into two segments AD and DC, the lengths where of may be found by theorem 4, thus:		
As the bafe AC	185 7	2.26893
To the form of the sides AB and BC 240	2.8034	
So is the difference of the files 72	1.85733	
To the diff, of the figures of the bafa 93	1.0687i	
And having the fum of the fragments, viz. the whole bafs, and their difference, we find the fragments themselves, by theorem 3, thus:		
To half the fum of the fragments	92E-6	
And half their difference	46-5	
And the fum is the greater fragment AD	139-7	
Allo from half the fum of the fragment	52-8	
Take half their difference	46-5	
The remainder is the lifter fragment DC	46-3	
Now the triangle ABC is divided, by the perpendicular DB, into two right angles triangles, ADB and DRG; in the lift of which are given the hypothecule AB=156, and the bafe AD=130-3, to find the oblique angles, for which we have (by Cafel V, of rechargular trigonometry) the fol-lowing analogy, viz.		
As AB	159	2.10312
Tier AD	193	2.1359
So is the radius	90°	1.00000
To the co-fise of the angle A	26° 40°	9.95083
Alfo the angle C is found by the fam cafe, thus:		
As IB 84	1.92428	
To CD	46:3	1.66518
So is the radius	90°	10.0000
To the co-fine of C	56° 30°	9.74130
Having found the two angles A and C, we have the third, B, by taking the time of the other two from 150, thus:		
The fum of all the three angles is 180°	The fum of A and C is	83° 10°
The angle B is	96° 50°	
All the proportions ufd for the foliations of the fever et cafes in plain trigonometry, may be performed by the final and compd. On the leaf there are several logarithmic lines, viz. one of numbers, another of fines, and one of tangents, éx.		
And the way of working a proportion by these is this, viz., extend your compaifles from the frit term of your pro portion, found on the feale, to the second; and with that extent, flying on foot in the third term, the other will reach the fourth term required.		

SPHERICAL TRIGONOMETRY

SPHERICAL TRIGONOMETRY is the art whereby, from three given parts of a spherical triangle, we discover the refl; and, like plane trigonometry, is either right-angled, or obliquely-angle]. But there we give the analogies for the foliation of the Several cases in other, it will be proper to premile the following theorems.

Taraxad. I. In all right-angled, spherical triangles, the fino of the hypothecial’s radius : fine of a leg - fine of its opposite angle. And the fine of a leg: radius : tangent of the other the tangent of its opposite angle.

Demonstration, Let E DA X (this id, fig. 1) re-prefent the eighth part of a Sphere, where the quadratil plane EDFG, DRG, can both prepened to the quadri-plane EDFs; and the quadratil plane ADG is perperendicular to the plane EDFG; and the spherical triangle ARC is right-angle as B, where CA is the hypo-theme, and BA, BC, so the legs.

To the arches GF, C B, draw the tangents HF, OB, and the finses CM G and the radii DP DB; and draw list the fine of the arch AB, and Ct the fine of AC; and then join IK and OL NowHF, OB, GM, CL are all perpendiculor to the plane ADDB. And HD, GK, OL, the all in the fans plane ADGC. Therefore, the right-angled triangles HHD, CKK, ODL, having the equal angles HDF, CKK OLB, see further. And CK-NOM CI COM; that is, as the fine of the hypothetic: rad:: fine of a leg - fine of its opposite angle. For OM is the fine of the arc GC, which therefore the angle CAB. Both, LB DF+; BO: FH; that is, as the fino of a leg: radius : tangent of the other leg: tangent of its opposite angle, J, E. D. Hence is follows: that the first of the angles of any ex-blique spherical triangle ACD (_this_, o^2), letto one anothey, directly, as the fins of the opposite fides. Hence it allo follows, that, in right-angledspherical triangles, having the fame perpendicular, the fino of the bafs will be to each other, inverify, as the tangents of the angles at the bafs.

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